

Extensive Form Games: Examples

An *extensive form game* is a collection $\Gamma = (N, A, H, \iota, \mathbb{I}, \{U_i\}_{i \in N})$.

1. Example 1.

- $N = \{1, 2\}$, $A = \{a, d, \ell, r\}$
- $H = \{h_0, h_0a, h_0d, h_0d\ell, h_0dr\}$
- $T = \{h_0a, h_0d\ell, h_0dr\}$
- $\iota(h_0) = 1$, $\iota(h_0d) = 2$
- $\mathbb{I} = \{\{h_0\}, \{h_0d\}\}$
- $(U_1(h_0a), U_2(h_0a)) = (0, 5)$, $(U_1(h_0d\ell), U_2(h_0d\ell)) = (-1, -1)$, $(U_1(h_0dr), U_2(h_0dr)) = (2, 2)$.

It is customary to omit h_0 in the terminal history as an argument of U_i and write instead $U_i(a)$, $U_i(d\ell)$, and $U_i(dr)$.

$$\mathbb{I}_1 = \{\{h_0\}\}, \mathbb{I}_2 = \{\{h_0d\}\}. S_1 = \{\{(h_0, a)\}, \{(h_0, d)\}\}, S_2 = \{\{(h_0d, \ell)\}, \{(h_0d, r)\}\}$$

It is customary to omit h_0 and write the strategy sets in the shorthand $S_1 = \{a, d\}$, $S_2 = \{\ell, r\}$. The utility representation of preferences on S are given by $(u_1(a, s_2), u_2(a, s_2)) = (0, 5)$, $(u_1(d, \ell), u_2(d, \ell)) = (-1, -1)$, $(u_1(d, r), u_2(d, r)) = (2, 2)$.

2. Stackelberg Duopoly Game

- $N = \{1, 2\}$, $A = \mathbb{R}_+$ (outputs)
- $H = \{h_0, \{h_0q_1\}_{q_1 \in \mathbb{R}_+}, \{h_0q_1q_2\}_{(q_1, q_2) \in \mathbb{R}_+^2}\}$
- $T = \{h_0q_1q_2\}_{(q_1, q_2) \in \mathbb{R}_+^2}$
- $\iota(h_0) = 1$, $\iota(h_0q_1) = 2$ for every $q_1 \geq 0$.
- $\mathbb{I} = \{\{h_0\}, \{\{h_0q_1\}\}_{q_1 \in \mathbb{R}_+}\}$
- $U_i(h_0q_1q_2) = p(q_1 + q_2)q_i - c_i(q_i)$ for $i = 1, 2$.

Again, it is customary to write $U_i(q_1, q_2)$ for $U_i(h_0q_1q_2)$.

$$\mathbb{I}_1 = \{\{h_0\}\}, \mathbb{I}_2 = \{\{h_0q_1\}\}_{q_1 \in \mathbb{R}_+}.$$

$$S_1 = \{(h_0, q_1)\}_{q_1 \in \mathbb{R}_+}. \text{ In short: } S_1 = \mathbb{R}_+.$$

For $f \subset A \times B$, write $\text{dom} f = \{a \in A \mid (a, b) \in f\}$. Then

$s_2 \in S_2$ if and only if $s_2 \subset \mathbb{R}_+ \times \mathbb{R}_+$, $\text{dom} s_2 = \mathbb{R}_+$, and “ $(q_1, q'_2) \in s_2$ & $(q_1, q_2) \in s_2$ ” implies $q_2 = q'_2$. In short: S_2 is the set of *functions* from $\mathbb{R}_+$ into $\mathbb{R}_+$.

The utility representations on S are given by

$$u_1(q_1, s_2(\cdot)) = p(q_1 + s_2(q_1))q_1 - c_1(q_1)$$

and

$$u_2(q_1, s_2(\cdot)) = p(q_1 + s_2(q_1))s_2(q_1) - c_2(s_2(q_1)).$$

where $(q_1, s_2) \in S$.